

Qi Xu

Spatial World Models ■ Embodied Intelligence ■ Multimodal Learning

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EDUCATION



Westlake University

PhD Student in Artificial Intelligence

Supervised by [Peidong Liu](#); Research: Spatial Intelligence, Embodied AI, Multimodal Learning

2026 – Present



Wuhan University

Master in Pattern Recognition and Intelligent Systems

Research: Computer Vision, 3D Reconstruction, Multimodal Learning

2023 – 2026



Wuhan University

B.S. in Spatial Information and Digital Technology

2019 – 2023

RESEARCH PROFILE

I study how intelligent agents **build, predict, and act on representations of the physical world**. My work spans 3D scene reconstruction and generation, long-horizon video reasoning, and multimodal post-training, building toward spatially grounded, action-aware world models for embodied agents.

EXPERIENCE



ByteDance, TikTok Global Live LLM Strategy Department

2025 – 2026

Research Intern; Mentored by [Xiangtai Li](#)

- Co-developed **video MLLMs for temporal grounding**, with ownership of SFT/RL post-training pipelines, reward design, and evaluation.
- Built scalable data and annotation pipelines for high-quality, temporally grounded multimodal training data.
- Scaled distributed training and evaluation to **hundreds of GPUs**; analyzed training stability and model convergence.
- Designed agentic workflows leveraging LLMs to automate real-time interaction analysis and optimize content distribution within the TikTok Live ecosystem.

SELECTED PUBLICATIONS



SIU3R: Simultaneous Scene Understanding and 3D Reconstruction Beyond Feature Alignment

NeurIPS 2025 Spotlight 160+ GitHub Stars 10+ Citations [Project Page](#) [ArXiv](#) [Code](#)

Qi Xu*, Dongxu Wei*, Lingzhe Zhao, Wenpu Li, Zhangchi Huang, Shunping Ji, Peidong Liu

Introduced a shared pixel-aligned 3D representation for simultaneous reconstruction and scene understanding from unposed images, providing a unified spatial state for geometry and semantics.

My Contributions: Project lead and primary code implementer; co-developed the core method and led benchmarking, ablation analysis, and paper writing.

2025



Any 3D Scene is Worth 1K Tokens: 3D-Grounded Representation for Scene Generation at Scale

arXiv 2026 [Project Page](#) [ArXiv](#) [Code](#)

Dongxu Wei*, Qi Xu*, Zhiqi Li, Hangning Zhou, Cong Qiu, Hailong Qin, Mu Yang, Zhaopeng Cui, Peidong Liu

Introduced a compact 1K-token implicit 3D latent representation and 3D Diffusion Transformer for scalable scene generation.

My Contributions: Co-designed the core method; implemented the training code and led model training; built the large-scale training dataset and led ablations and evaluation.

2026



Towards One-to-Many Temporal Grounding

ICML 2026 [Project Page](#) [ArXiv](#)

Qi Xu*, Yue Tan*, Shihao Chen, Jiahao Meng, Anran Wang, Shunping Ji, Hao Fei, Xiangtai Li

Introduced the first systematic solution for OMTG, featuring a 56k-sample dataset and RL training with CoT-based rewards; outperformed Gemini 2.5 Pro by 15.85%.

My Contributions: Co-initiated the project and co-developed the core idea; owned data construction, SFT/RL pipelines, reward design, model training, benchmarking, and paper writing.

2026

ADDITIONAL PUBLICATIONS



VideoZeroBench: Probing the Limits of Video MLLMs with Spatio-Temporal Evidence Verification

arXiv 2026 [Project Page](#) [ArXiv](#) [Code](#)

Jiahao Meng, Tan Yue, Qi Xu, Haochen Wang, Zhongwei Ren, Weisong Liu, Yuhao Wang, Renrui Zhang, Yunhai Tong, Haodong Duan

2026

A hierarchical long-video QA benchmark that verifies model answers with temporal intervals and spatial boxes under a five-level evidence protocol.

My Contributions: Co-designed the five-level spatio-temporal evidence evaluation protocol; designed and implemented the annotation tool and guidelines; constructed approximately 200 complete QA-evidence samples.



Watch, Remember, Reason: Human-View Video Understanding with MLLMs

arXiv 2026 [ArXiv](#) [GitHub](#)

Jiahao Meng, Yue Tan, Qi Xu, Kuan Gao, Weisong Liu, Yanwei Li, Xiangtai Li, Lingdong Kong, Haochen Wang, Qianyu Zhou, Jiangning Zhang, Guangliang Cheng, Yunhai Tong, Lu Qi, Minghsuan Yang

2026

Surveyed human-view video understanding with MLLMs through watching, remembering, and reasoning abilities for long, multimodal, and knowledge-intensive video scenarios.

My Contributions: organized literature for the Memory section, created figures and tables, and contributed to paper writing and revision.



HiCI: Hierarchical Construction-Integration for Long-Context Attention

ICML 2026 [ArXiv](#) [Code](#)

Xiangyu Zeng, Qi Xu, Yunke Wang, Chang Xu

2026

Efficient long-context modeling using hierarchical attention, extending LLaMA-2 to 100K tokens with only 5.5% additional parameters.

My Contributions: participated in partial method design and paper revision.



E-MoFlow: Learning Egomotion and Optical Flow from Event Data via Implicit Regularization

NeurIPS 2025 [Project Page](#) [ArXiv](#) [Code](#)

Wenpu Li, Bangyan Liao, Yi Zhou, Qi Xu, Pian Wan, Peidong Liu

2025

An unsupervised framework for joint ego-motion and optical flow estimation using implicit neural representations and differential geometric constraints.

My Contributions: participated in partial method design and paper revision.

INVITED TALKS

Post-Training of Multimodal Large Language Models

Aug 6, 2026

Invited Talk, Hangzhou Dianzi University

Data, Optimization and Evaluation [Slides](#)

RESEARCH MENTORSHIP



SupIR-GS: Thermal Infrared Super-Resolution Novel View Synthesis with Imaging-Calibrated 3DGS

ECCV 2026 **Corresponding Author**

Jin Liu, Haodong Li, Jiagang Chen, Dabin leng, Jiguang Li, Zhao Huang, Xiaoshuai Zhang, Qi Xu[†], Zhiwen Zheng,

2026

Xingru Huang[†]

Reconstructs high-resolution 3D thermal scenes from low-resolution inputs via physics-informed degradation modeling.

Mentorship: Provided the core idea, guided method design and experiment planning, and led paper writing and revision.



Physically Grounded Dual-Opacity Gaussian Splatting for Joint RGB-TIR Reconstruction

ECCV 2026 **Corresponding Author**

Jin Liu, Dabin leng, Jiagang Chen, Haodong Li, Jiguang Li, Zhao Huang, Xiaoshuai Zhang, Zhiwen Zheng, Xingru

2026

Huang[†], Qi Xu[†]

A unified RGB-TIR reconstruction framework resolving cross-spectral visibility conflicts through thermal field modeling.

Mentorship: Proposed the core idea, coordinated experiments and ablations, and led paper writing and revision.

SKILLS

Research: 3D reconstruction and generation, video reasoning, spatio-temporal grounding, multimodal SFT/RL post-training, benchmark construction, and test-time scaling.

Direction: Action-conditioned world models, physical reasoning, planning, and learning for embodied agents.

Engineering: Python, PyTorch, distributed training/evaluation, data construction pipelines, and large-scale experimental analysis; familiar with C++ and Linux development.

Writing & Leadership: Leading research projects from problem formulation to implementation, benchmarking, analysis, paper writing, and public release.